# TRIGONOMETRY

## Trigonometry Ratio

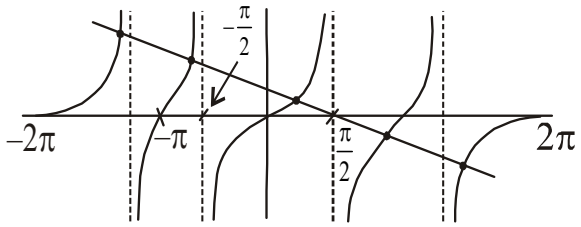
1. The number of solutions of the equation $2x + 3\tan x = \pi,\ x \in [-2\pi, 2\pi] - \left\{\pm\frac{\pi}{2}, \pm\frac{3\pi}{2}\right\}$ is :

समीकरण $2x + 3\tan x = \pi,\ x \in [-2\pi, 2\pi] - \left\{\pm\frac{\pi}{2}, \pm\frac{3\pi}{2}\right\}$ के हलों की संख्या है :

(A) 6 (B) 5 (C) 3 (D) 4

**Ans.** B

**Sol.** $\tan x = \frac{\pi}{3} - \frac{2x}{3}$




2. If $10\sin^4\theta + 15\cos^4\theta = 6$, then the value of $\frac{27\,\text{cosec}^6\,\theta + 8\sec^6\theta}{16\sec^8\theta}$ is :

यदि $10\sin^4\theta + 15\cos^4\theta = 6$ है, तो $\frac{27\,\text{cosec}^6\,\theta + 8\sec^6\theta}{16\sec^8\theta}$ का मान है :

(A) $\frac{3}{4}$ (B) $\frac{2}{5}$ (C) $\frac{1}{5}$ (D) $\frac{3}{5}$

**Ans.** B

**Sol.** $10\sin^4\theta + 15\cos^4\theta = 6$

$\frac{27\,\text{cosec}^6\theta + 8\sec^6\theta}{16\sec^8\theta} = ?$

$10\tan^4\theta + 15 = 6\sec^4\theta$

$= 6(\sec^2\theta)^2$

$= 6(1 + \tan^2\theta)^2$

$= 6(1 + \tan^4\theta + 2\tan^2\theta)$

$4\tan^4\theta - 12\tan^2\theta + 9 = 0$

$(2\tan^2\theta - 3)^2 = 0$

$\tan^2\theta = \frac{3}{2}$

$\sec^2\theta = 1 + \tan^2\theta$

$= 1 + \frac{3}{2}$

$\sec^2\theta = \frac{5}{2}$

& $\cot^2\theta = \frac{2}{3}$

$\Rightarrow \operatorname{cosec}^2\theta = 1 + \frac{2}{3} = \frac{5}{3}$

Now, $\frac{27\left(\operatorname{cosec}^2\theta\right)^3 + 8\left(\sec^2\theta\right)^3}{16\left(\sec^2\theta\right)^4}$

$\frac{27\left(\frac{5}{3}\right)^3 + 8\left(\frac{5}{2}\right)^3}{16\left(\frac{5}{2}\right)^4}$

$\frac{5^3 + 5^3}{5^4} \Rightarrow \frac{2.5^3}{5^4}$

$\Rightarrow \frac{2}{5}$

3. Let $\alpha_\theta$ and $\beta_\theta$ be the distinct roots of $2x^2 + (\cos\theta)x - 1 = 0$, $\theta \in (0, 2\pi)$. if m and M are the minimum and the maximum values of $\alpha_\theta^4 + \beta_\theta^4$, then 16(M + m) equals :

माना कि $\alpha_\theta$ और $\beta_\theta$, $2x^2 + (\cos\theta)x - 1 = 0$, $\theta \in (0, 2\pi)$ के भिन्न मूल हैं। यदि m और M, $\alpha_\theta^4 + \beta_\theta^4$ के न्यूनतम और उच्चतम मान हैं, तो 16(M + m) बराबर है :

(A) 17 (B) 24 (C) 27 (D) 25

**Ans.** D

**Sol.** $2x^2 + (\cos\theta)x - 1 = 0$ $\theta \in (0, 2\pi)$

$\alpha_\theta + \beta_\theta = -\frac{\cos\theta}{2}$ $\alpha_\theta \cdot \beta_\theta = -\frac{1}{2}$

Now $= \alpha_\theta^4 + \beta_\theta^4$

$= \left(\alpha_\theta^2 + \beta_\theta^2\right)^2 - 2\alpha_\theta^2\beta_\theta^2$

$$= \left\{ \left( \alpha_\theta + \beta_\theta \right)^2 - 2\alpha_\theta \cdot \beta_\theta \right\}^2 - 2\alpha_\theta^2 \beta_\theta^2$$

$$= \left\{ \frac{\cos^2 \theta}{4} + 1 \right\}^2 - 2 \times \frac{1}{4}$$

$$= \left( 1 + \frac{1}{4} \cos^2 \theta \right)^2 - \frac{1}{2}$$

[0, 1]

Min value when $\cos^2\theta = 0$

$$m = \frac{1}{2}$$

Max. value when $\cos^2\theta = 1$

$$\Rightarrow \left( 1 + \frac{1}{4} \right)^2 - \frac{1}{2} = \frac{25}{16} - \frac{1}{2} = \frac{17}{16}$$

$$M = \frac{17}{16}$$

Now $16(M + m)$

$$\Rightarrow 16\left( \frac{17}{16} + \frac{1}{2} \right) = 25$$

4. The value of $(\sin 70^\circ)(\cot 10^\circ \cot 70^\circ - 1)$ is :

$(\sin 70^\circ)(\cot 10^\circ \cot 70^\circ - 1)$ का मान है :

(A) 2/3 (B) 3/2 (C) 1 (D) 0

**Ans.** C

**Sol.** $$\sin 70^\circ \left( \frac{\cos 10^\circ \cos 70^\circ - \sin 10^\circ \sin 70^\circ}{\sin 10^\circ \sin 70^\circ} \right)$$

$$= \frac{\cos(80)}{\sin 10^\circ} = 1$$

5. If $\sum_{r=1}^{13} \left\{ \frac{1}{\sin\left( \frac{\pi}{4} + (r-1)\frac{\pi}{6} \right) \sin\left( \frac{\pi}{4} + \frac{r\pi}{6} \right)} \right\} = a\sqrt{3} + b$ , a, b $\in$ Z, then $a^2 + b^2$ is equal to :

यदि $\sum_{r=1}^{13}\left\{\frac{1}{\sin\left(\frac{\pi}{4}+(r-1)\frac{\pi}{6}\right)\sin\left(\frac{\pi}{4}+\frac{r\pi}{6}\right)}\right\}=a\sqrt{3}+b$ , a, b $\in$ Z, तो $a^2+b^2$ है, तो बराबर है :

(A) 2 (B) 4 (C) 8 (D) 10

**Ans.** C

**Sol.** $\sum_{r=1}^{13}\left[\frac{1}{\sin\left(\frac{\pi}{4}+(r-1)\frac{\pi}{6}\right)\sin\left(\frac{\pi}{4}+\frac{r\pi}{6}\right)}\right]$

$$\frac{1}{\left(\sin\left(\frac{\pi}{6}\right)\right)}\sum_{r=1}^{13}\left[\frac{\sin\left(\left(\frac{\pi}{4}+\frac{r\pi}{6}\right)-\left(\frac{\pi}{4}+\frac{(r-1)\pi}{6}\right)\right)}{\sin\left(\frac{\pi}{4}+(r-1)\frac{\pi}{6}\right)\sin\left(\frac{\pi}{4}+\frac{r\pi}{6}\right)}\right]$$

$$\frac{1}{\sin\left(\frac{\pi}{6}\right)}\sum_{r=1}^{13}\left[\cot\left(\frac{\pi}{4}+\frac{(r-1)\pi}{6}\right)-\cot\left(\frac{\pi}{4}+\frac{r\pi}{6}\right)\right]$$

$$(2)\left[\left(\cot\left(\frac{\pi}{4}\right)-\cot\left(\frac{\pi}{4}+\frac{\pi}{6}\right)\right)+\left(\cot\left(\frac{\pi}{4}+\frac{\pi}{6}\right)-\cot\left(\frac{\pi}{4}+\frac{2\pi}{6}\right)\right)+....\cot\left(\frac{\pi}{4}+\frac{12\pi}{6}\right)-\cot\left(\frac{\pi}{4}+\frac{13\pi}{6}\right)\right]$$

$$(2)\left[\cot\left(\frac{\pi}{4}\right)-\cot\left(\frac{\pi}{4}+2\pi+\frac{\pi}{6}\right)\right]$$

(B) [1 – cot (75°)]

$$(2)\left(1-\left(2-\sqrt{3}\right)\right)$$

$$(2)\left(\sqrt{3}-1\right)$$

$$\left(2\sqrt{3}-2\right)\Rightarrow a\sqrt{3}+b \Rightarrow a=2, b=-2$$

$a^2 + b^2 = 4 + 4 = 8$

6. If $\sin x+\sin^2 x=1, x\in\left(0,\frac{\pi}{2}\right)$, then

$\left(\cos^{12} x+\tan^{12} x\right)+3\left(\cos^{10} x+\tan^{10} x+\cos^8 x+\tan^8 x\right)+\left(\cos^6 x+\tan^6 x\right)$ is equal to :

यदि $\sin x + \sin^2 x = 1, x \in \left(0, \frac{\pi}{2}\right)$ है, तो

$\left(\cos^{12} x + \tan^{12} x\right) + 3\left(\cos^{10} x + \tan^{10} x + \cos^8 x + \tan^8 x\right) + \left(\cos^6 x + \tan^6 x\right)$ बराबर है :

(A) 4 (B) 2 (C) 3 (D) 1

**Ans.** B

**Sol.** $\sin x + \sin^2 x = 1$

$\Rightarrow \sin x = \cos^2 x \qquad \Rightarrow \tan x = \cos x$

$\therefore$ Given expression

$= 2\cos^{12} x + 6[\cos^{10} x + \cos^8 x] + 2\cos^6 x$

$= 2[\sin^6 x + 3\sin^5 x + 3\sin^4 x + \sin^3 x]$

$= 2\sin^3 x[(\sin x + 1)^3]$

$= 2[\sin^2 x + \sin x]^3$

$= 2$

7. Let $\left|\cos\theta \cdot \cos(60 - \theta) \cdot \cos(60 + \theta)\right| \le \frac{1}{8}, \theta \in [0, 2\pi]$. Then, then sum of all $\theta \in [0, 2\pi]$, where $\cos 3\theta$ attains its maximum value of is :

माना $\left|\cos\theta \cdot \cos(60 - \theta) \cdot \cos(60 + \theta)\right| \le \frac{1}{8}, \theta \in [0, 2\pi]$ है। तो $\theta \in [0, 2\pi]$ के सभी मानों, जिनके लिए $\cos 3\theta$ का मान उच्चतम है, का योग है :

(A) $18\pi$ (B) $9\pi$ (C) $15\pi$ (D) $6\pi$

**Ans.** D

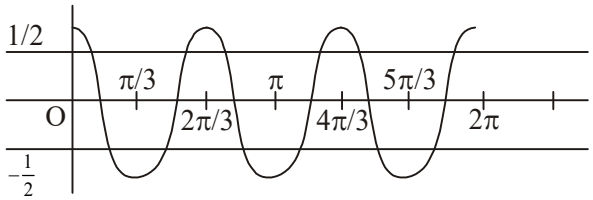
**Sol.** $\cos\theta.\cos(60 - \theta)\cos(60 + \theta) = \frac{1}{4}\cos 3\theta$

$\therefore |\cos 3\theta| \le \frac{1}{2}$

$0 \le 3\theta \le 6\pi$

$-\frac{1}{2} \le \cos 3\theta \le \frac{1}{2}$

$\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{9\pi}{6}, \frac{11\pi}{6}$




θ satisfying condition are $\frac{\pi}{9},\frac{\pi}{2}+\frac{\pi}{18}+\frac{5\pi}{6}-\frac{\pi}{18},\frac{7\pi}{6}+\frac{\pi}{16},\frac{3\pi}{2}-\frac{\pi}{18},\frac{11\pi}{6}+\frac{\pi}{18}$

$$\sum\theta=\frac{\pi}{9},\frac{\pi}{2}+\frac{\pi}{6}+\frac{7\pi}{6}+\frac{3\pi}{2}+\frac{11\pi}{6}+\frac{\pi}{18}$$

$$=\frac{(2+9+15+21+27+33+1)\pi}{18}=6\pi$$

8. If $\tan A=\frac{1}{\sqrt{x\left(x^2+x+1\right)}}$, $\tan B=\frac{\sqrt{x}}{\sqrt{x^2+x+1}}$ and $\tan C=\left(x^{-3}+x^{-2}+x^{-1}\right)^{\frac{1}{2}}$, $0<A,B,C<\frac{\pi}{2}$, then A + B is equal to :

यदि $\tan A=\frac{1}{\sqrt{x\left(x^2+x+1\right)}}$, $\tan B=\frac{\sqrt{x}}{\sqrt{x^2+x+1}}$ तथा $\tan C=\left(x^{-3}+x^{-2}+x^{-1}\right)^{\frac{1}{2}}$, $0<A,B,C<\frac{\pi}{2}$ हैं, तो A + B बराबर है :

(A) $\pi - C$ (B) $\frac{\pi}{2}-C$ (C) $2\pi - C$ (D) C

**Ans.** D

**Sol.** $\tan A=\frac{1}{\sqrt{x\left(x^2+x+1\right)}},\tan B=\frac{\sqrt{x}}{\sqrt{x^2+x+1}}$

$$\tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\cdot\tan B}$$

$$\tan(A+B)=\frac{\frac{1}{\sqrt{x\left(x^2+x+1\right)}}+\frac{\sqrt{x}}{\sqrt{x^2+x+1}}}{1-\frac{1}{\sqrt{x\left(x^2+x+1\right)}}\cdot\frac{\sqrt{x}}{\sqrt{x^2+x+1}}}$$

$$\tan(A+B)=\frac{1+x}{\sqrt{x}\cdot\sqrt{x^2+x+1}}\times\frac{\left(x^2+x+1\right)}{x^2+x}$$

$$\tan(A+B)=\frac{\sqrt{x^2+x+1}}{x\sqrt{x}} \qquad ...(1)$$

Given $\tan C=\left(\frac{1}{x^3}+\frac{1}{x^2}+\frac{1}{x}\right)^{\frac{1}{2}}$

$$\tan C=\frac{\sqrt{1+x+x^2}}{x\sqrt{x}} \qquad ...(2)$$

from equation (1) & (2)

$\tan(A+B)=\tan C$

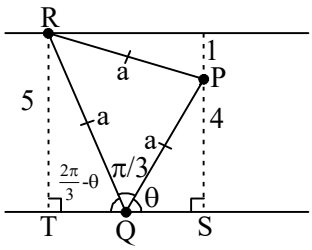
or $A+B=C$

9. Let the distance between two parallel lines be 5 units and a point P lie between the lines at a unit distance from one of them. An equilateral triangle PQR is formed such that Q lies on one of the parallel lines, while R lies on the other. Then $(Q\,R)^2$ is equal to _____.

माना कि दो समान्तर रेखाओं के बीच की दूरी 5 इकाई है और एक बिन्दु P रेखाओं के बीच उनमें से एक से 1 इकाई की दूरी पर स्थित है। एक समबाहु त्रिभुज PQR इस प्रकार बनाया जाता है कि Q समान्तर रेखाओं में से एक पर जबकि R दूसरी पर स्थित हैं तब $(Q\,R)^2$ बराबर है _____।

**Ans.** 28

**Sol.**




In $\Delta PQS$ $\qquad \sin\theta=\frac{4}{a}$

In $\Delta QRT$ $\qquad \sin\left(\frac{2\pi}{3}-\theta\right)=\frac{5}{a}$

$$\sin\frac{2\pi}{3}\cos\theta-\cos\frac{2\pi}{3}\sin\theta=\frac{5}{a}$$

$$\frac{\sqrt{3}}{2}\times\frac{\sqrt{a^2-16}}{a}+\frac{1}{2}\times\frac{4}{a}=\frac{5}{a}$$

$$\sqrt{3}\sqrt{a^2-16}+4=10$$

$a^2 - 16 = 12$

$a^2 = 28$

$(QR)^2 = 28$

10. If the range of $f(\theta) = \frac{\sin^4\theta + 3\cos^2\theta}{\sin^4\theta + \cos^2\theta}, \theta \in R$ is $[\alpha, \beta]$, then the sum of the infinite G.P., whose first term is 64 and the common ratio is $\frac{\alpha}{\beta}$, is equal to _____.

यदि $f(\theta) = \frac{\sin^4\theta + 3\cos^2\theta}{\sin^4\theta + \cos^2\theta}, \theta \in R$ का परिसर $[\alpha, \beta]$ है, तब अनंत गुणोत्तर श्रेणी का योग क्या होगा, जिसका प्रथम पद 64 तथा सार्वनुपात $\frac{\alpha}{\beta}$ है, _____

**Ans.** 96

**Sol.** $f(\theta) = \frac{\sin^4\theta + 3\cos^2\theta}{\sin^4\theta + \cos^2\theta}$

$f(\theta) = 1 + \frac{2\cos^2\theta}{\sin^4\theta + \cos^2\theta}$

$f(\theta) = \frac{2\cos^2\theta}{\cos^4\theta - \cos^2\theta + 1} + 1$

$f(\theta) = \frac{2}{\cos^2\theta + \sec^2\theta - 1} + 1$

$f(\theta)|_{min} = 1$

$f(\theta)_{max} = 3$

$S = \frac{64}{1 - 1/3} = 96$